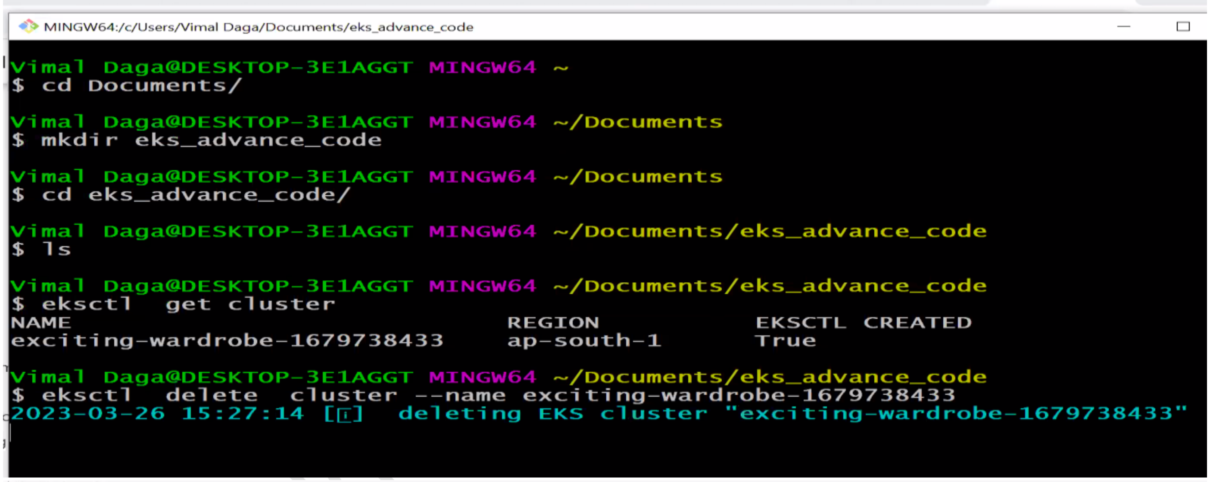
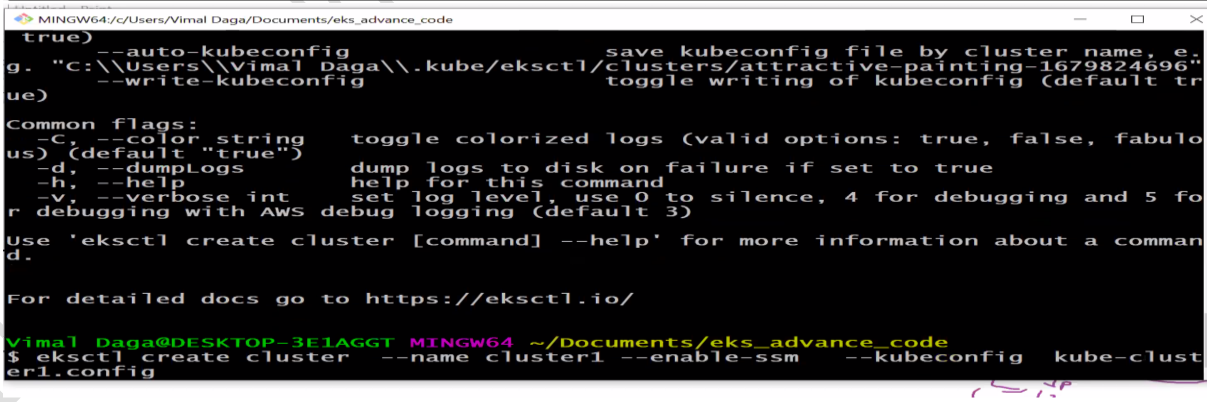
AWS Session

Summary – 26-03-2023

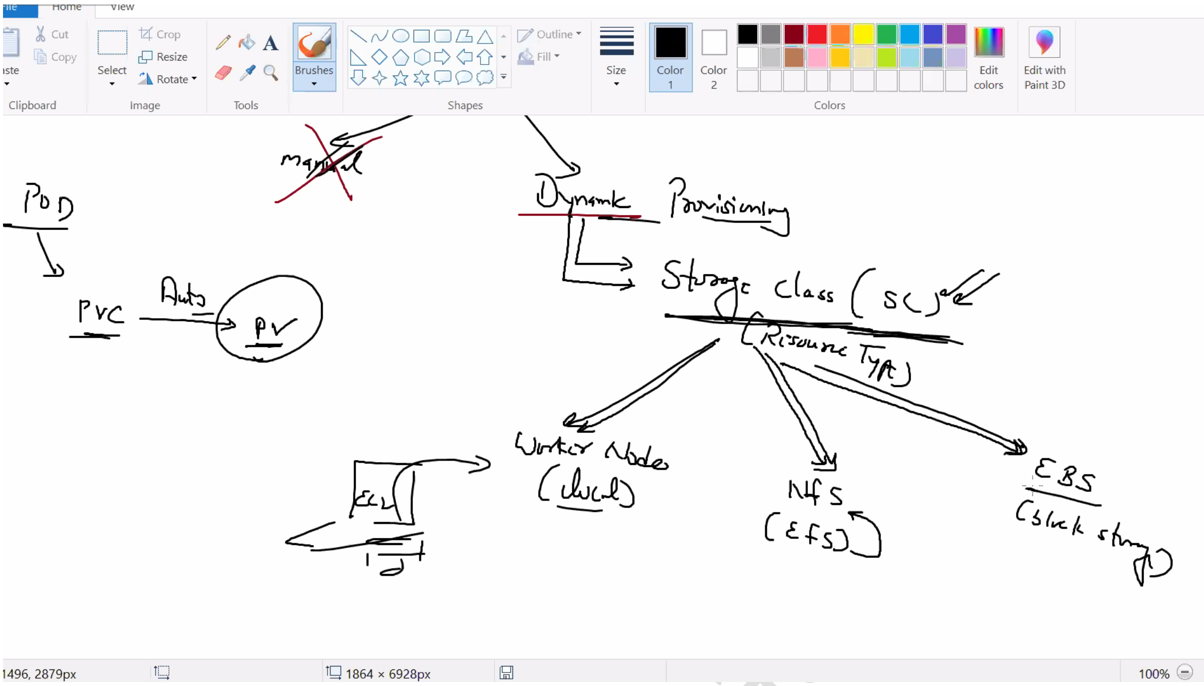
- Every cluster has some configuration file called **kubeconfig**.
- **kubectl** is a command for Kubernetes not for eksctl.
- Whenever we create a pod, one of the responsibilities of pod is to launch an app.





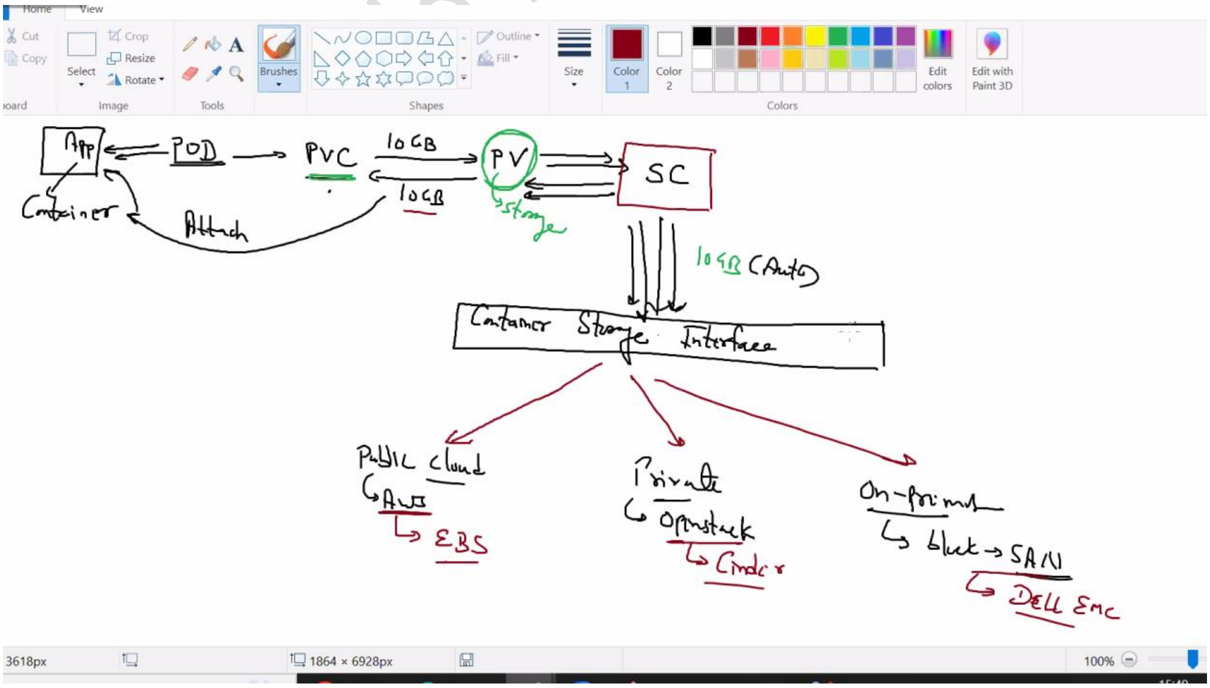
- Application needs storage(permanent/persistent). It is the duty of pod to claim(request) for persistent volume(PV), this concept is called **Persistent volume claim(PVC).**
- 2 ways to create Persistent Volume (Storage)
  1. Manually
  2. Dynamically

- SC – Storage class help to provision storage dynamically. SC is a resource type.



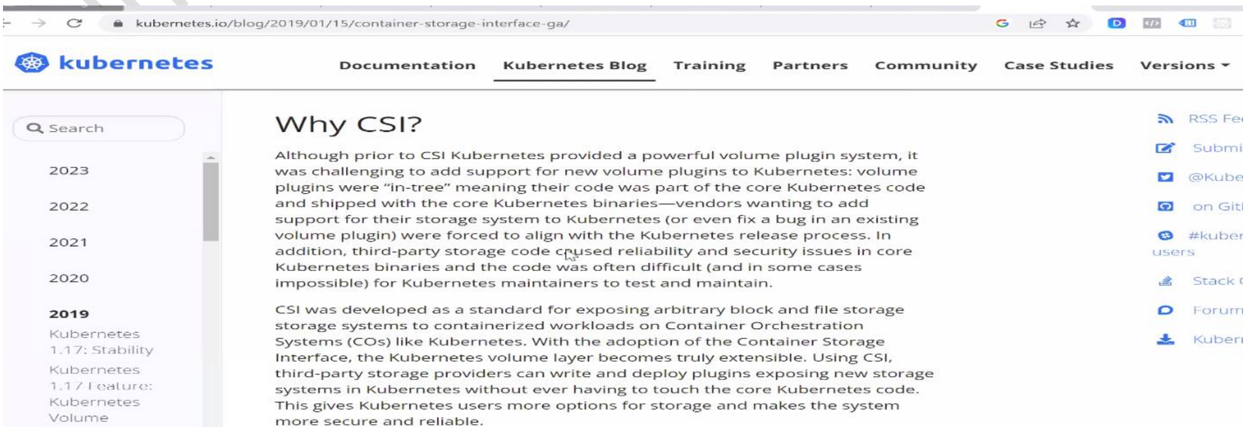
- Pod is the one who is responsible to run the app.
- Storage class can take storage from:

1. Public Cloud (AWS- EBS)
2. Private Cloud (Openstack- Cinder)
3. On-premises (block-SAN- dell, EMC)



- In k8s, there is one standard interface for the storage. So, if container storage wants to connect with EBS then we need to give a driver/provisioner, So in AWS that driver is known as **CSI Provisioner**.
- Kubernetes only understand PVC, PV, SC If we want Kubernetes resource type take from third party and for that we use CSI.
- Storage class is one that use driver from external provisioners.
- If we run our cluster on k8s with EKS, it will launch K8S and whenever EKS want storage it will connect to storage class (internally Integrated with other services of AWS like EBS, EFS)
- If we want to connect to a cluster then the credentials can be found in kube config file.
- PV is for storage and SC is storage class.
- As soon as we launch pod in EKS and pod need a storage, internally EKS connect with EBS service and launch a new volume and everything is possible because of driver/provisioner.
- Every resource type has annotations which give different responsibilities. In annotations there are keywords like parameters. Annotation is used in all resource type.
- SC need driver called CSI driver. It is a program which runs inside the pod in k8s.
- All internal drivers of k8s is managed in area called **namespace**. One of the namespace is kube-system.
- Whenever EKS launch, it creates role and attach to EKS.
- Configmap is a resource which like a document, maintain config file or maintain file in k8s.
- 2 ways to install driver are
  1. Addons (drivers)
  2. Self-managed addon (some kind of script like k8s scripts)

  Standard way is Drivers that is using CSI



PRACTICAL

MINGW64:/c/Users/Vimal Daga/Documents/eks_advance_code

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ pwd
/c/Users/Vimal Daga/Documents/eks_advance_code

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ ls
kube-cluster1.config
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ eksctl  get cluster
NAME                            REGION          EKSCTL CREATED
cluster1                        ap-south-1      True
exciting-wardrobe-1679738433    ap-south-1      True

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ ls
kube-cluster1.config
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl  get pods
E0326 16:01:19.625059   14756 memcache.go:238] couldn't get current server API g
roup list: Get "http://localhost:8080/api?timeout=32s": dial tcp [::1]:8080: con
nectex: No connection could be made because the target machine actively refused
it.
E0326 16:01:21.972382   14756 memcache.go:238] couldn't get current server API g
roup list: Get "http://localhost:8080/api?timeout=32s": dial tcp [::1]:8080: con
nectex: No connection could be made because the target machine actively refused
it.
E0326 16:01:24.328932   14756 memcache.go:238] couldn't get current server API g
roup list: Get "http://localhost:8080/api?timeout=32s": dial tcp [::1]:8080: con
nectex: No connection could be made because the target machine actively refused
it.
E0326 16:01:26.692897   14756 memcache.go:238] couldn't get current server API g
roup list: Get "http://localhost:8080/api?timeout=32s": dial tcp [::1]:8080: con
nectex: No connection could be made because the target machine actively refused
it.
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl  get  pvc  --kubeconfig  kube-cluster1.config
No resources found in default namespace.

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$
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl  get  pv  --kubeconfig  kube-cluster1.config
No resources found

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl  get  sc  --kubeconfig  kube-cluster1.config
NAME            PROVISIONER             RECLAIMPOLICY   VOLUMEBINDINGMODE
LLOWVOLUMEEXPANSION   AGE
gp2 (default)   kubernetes.io/aws-ebs   Delete          WaitForFirstConsumer
alse                  24m

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$
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl  describe   sc  gp2  --kubeconfig  kube-cluster1.config
Name:            gp2
IsDefaultClass:  Yes
Annotations:     kubectl.kubernetes.io/last-applied-configuration={"apiVersion":"storage.k8s.io/v1"
,"kind":"StorageClass","metadata":{"annotations":{"storageclass.kubernetes.io/is-default-class":"tr
ue"},"name":"gp2"},"parameters":{"fsType":"ext4","type":"gp2"},"provisioner":"kubernetes.io/aws-ebs
","volumeBindingMode":"WaitForFirstConsumer"}
,storageclass.kubernetes.io/is-default-class=true
Provisioner:           kubernetes.io/aws-ebs
Parameters:            fsType=ext4,type=gp2
AllowVolumeExpansion:  <unset>
MountOptions:          <none>
ReclaimPolicy:         Delete
VolumeBindingMode:     WaitForFirstConsumer
Events:                <none>

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$
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl   get pods    --kubeconfig  kube-cluster1.config
No resources found in default namespace.

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl   get pods    -n kube-system    --kubeconfig  kube-cluster1.config
NAME                       READY   STATUS    RESTARTS   AGE
aws-node-418xs             1/1     Running   0          28m
aws-node-qqg7w             1/1     Running   0          28m
coredns-74ccfbd98b-45w5p   1/1     Running   0          36m
coredns-74ccfbd98b-5826h   1/1     Running   0          36m
kube-proxy-6jcll           1/1     Running   0          28m
kube-proxy-tkbfd           1/1     Running   0          28m

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl  get  sc  --kubeconfig  kube-cluster1.config
NAME            PROVISIONER             RECLAIMPOLICY   VOLUMEBINDINGMODE      ALLOWVOLUMEEXPANSIO
   AGE
gp2 (default)   kubernetes.io/aws-ebs   Delete          WaitForFirstConsumer   false
   38m

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$
```

Gp2 is a storage class created by default.

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl   get  configmap    -n kube-system    --kubeconfig  kube-cluster1.config
NAME                                 DATA   AGE
aws-auth                             1      37m
coredns                              1      44m
cp-vpc-resource-controller           0      44m
eks-certificates-controller          0      44m
extension-apiserver-authentication   6      44m
kube-proxy                           1      44m
kube-proxy-config                    1      44m
kube-root-ca.crt                     1      44m

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl   get  configmap   aws-auth  -n kube-system    --kubeconfig  kube-cluster1.config
NAME       DATA   AGE
aws-auth   1      37m

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$
```
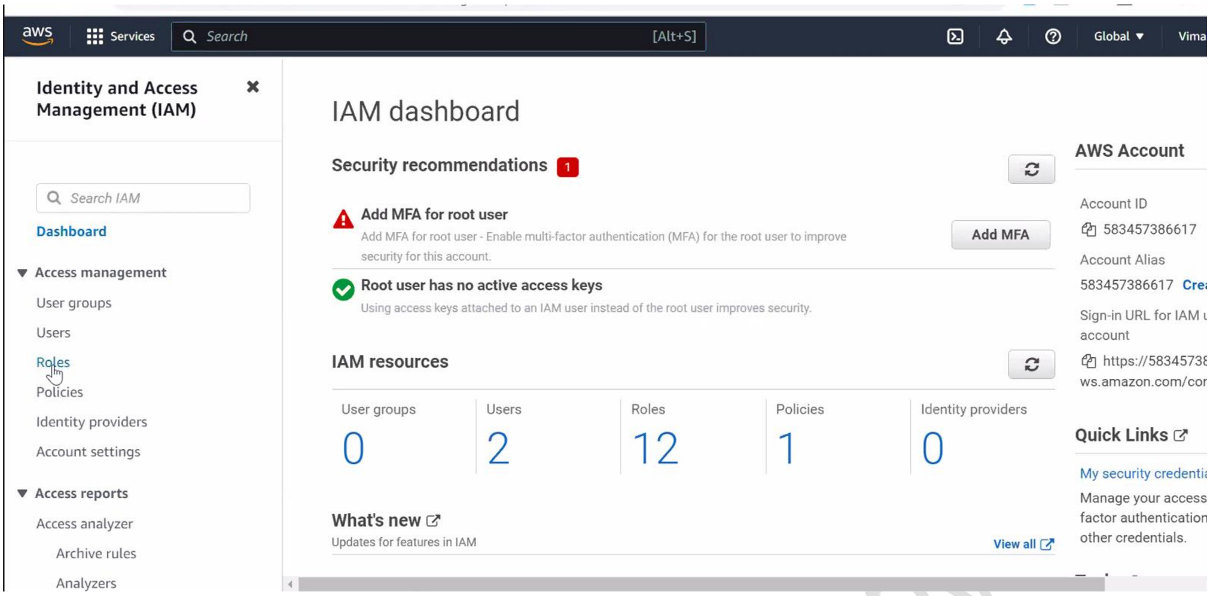
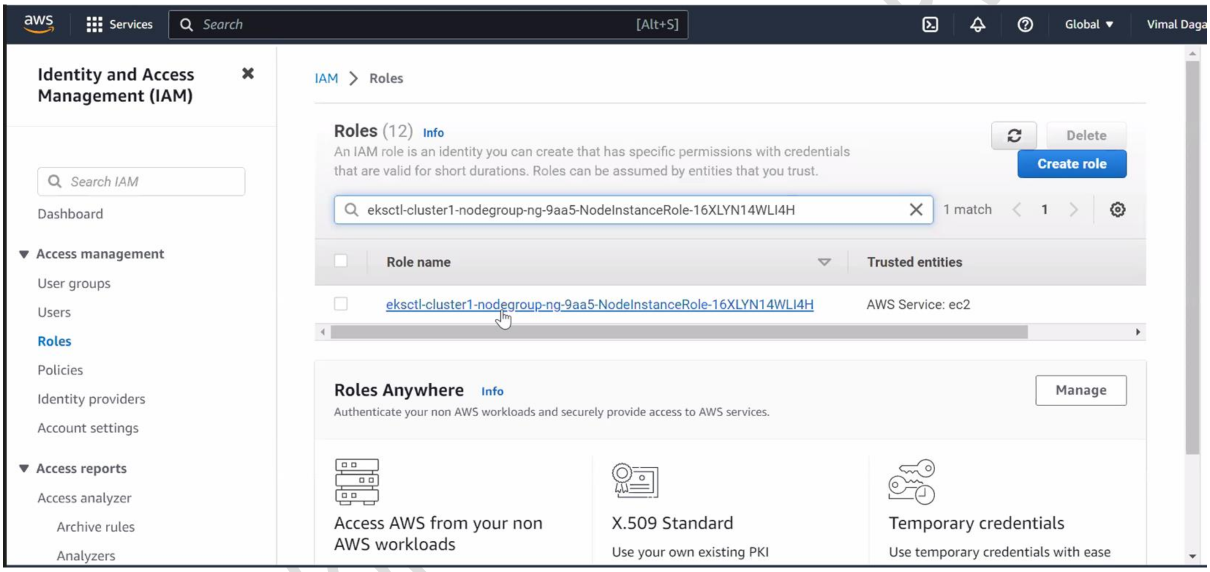

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl   get  configmap   aws-auth  -n kube-system    --kubeconfig  kube-cluster1.config
NAME       DATA   AGE
aws-auth   1      37m

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl   describe  configmap   aws-auth  -n kube-system    --kubeconfig  kube-cluster1.config
```

Now open IAM and create a role by attaching a permission policy.

[AWS]



aws
Services
Search
[Alt+S]
Global
Identity and Access Management (IAM)
Search IAM
Dashboard
Access management
User groups
Users
Roles
Policies
Identity providers
Account settings
Access reports
Access analyzer
Archive rules
Analyzers
IAM dashboard
Security recommendations 1
Add MFA for root user
Add MFA for root user - Enable multi-factor authentication (MFA) for the root user to improve security for this account.
Add MFA
Root user has no active access keys
Using access keys attached to an IAM user instead of the root user improves security.
IAM resources
User groups 0
Users 2
Roles 12
Policies 1
Identity providers 0
What's new
Updates for features in IAM
View all
AWS Account
Account ID
583457386617
Account Alias
583457386617
Sign-in URL for IAM users in this account
Quick Links
Manage your access keys, multi-factor authentication and other credentials.



aws
Services
Search
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Global
Identity and Access Management (IAM)
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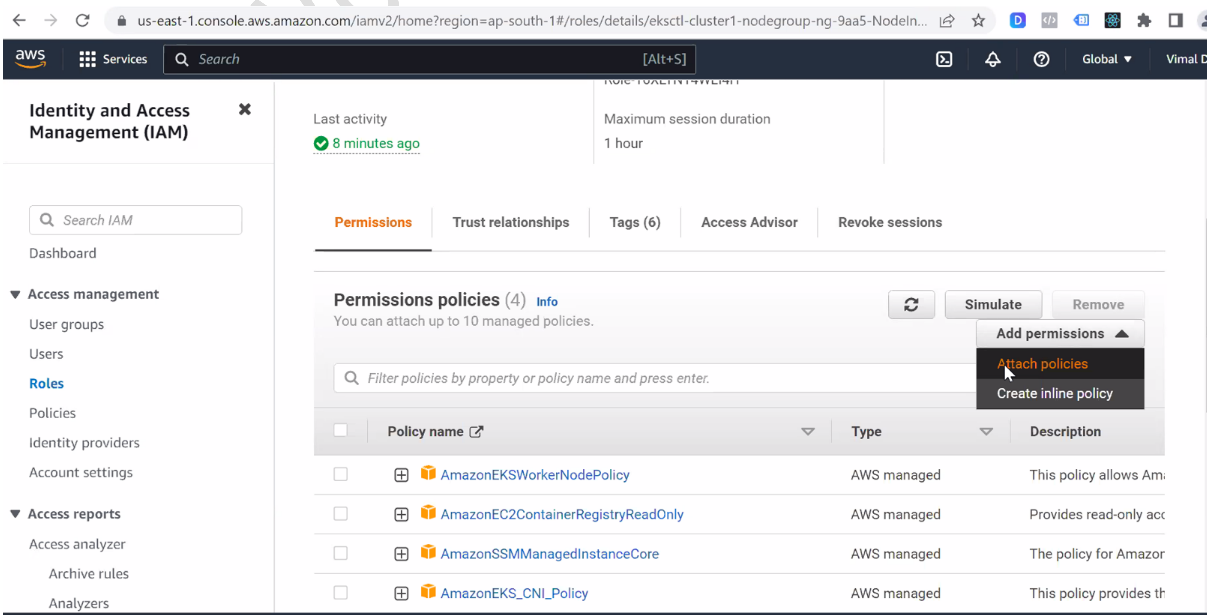
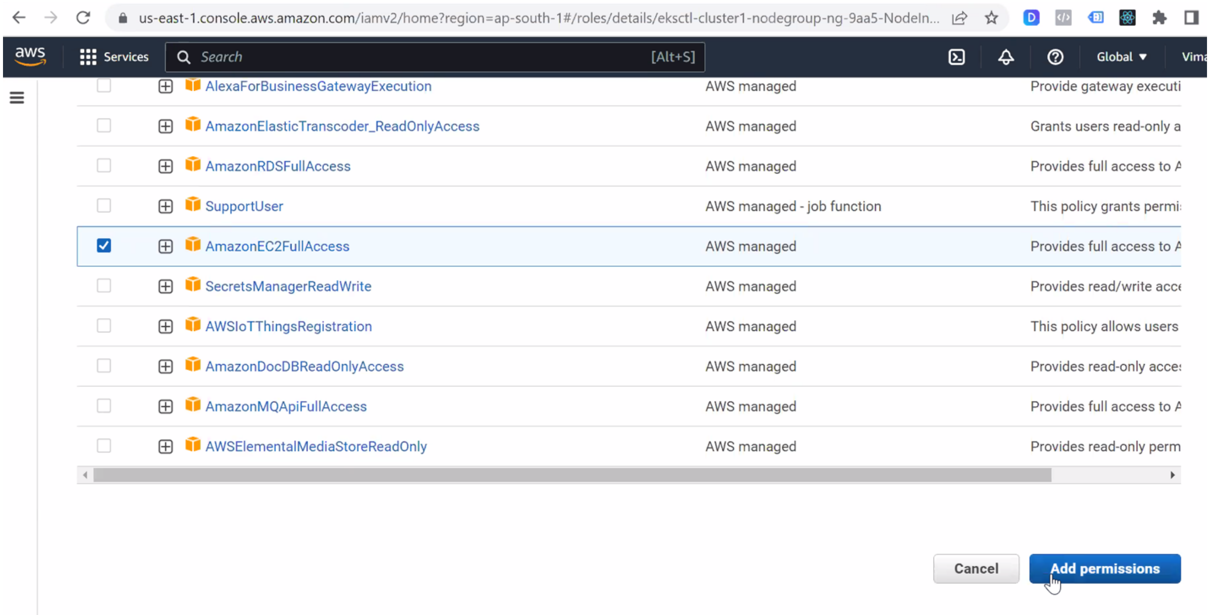
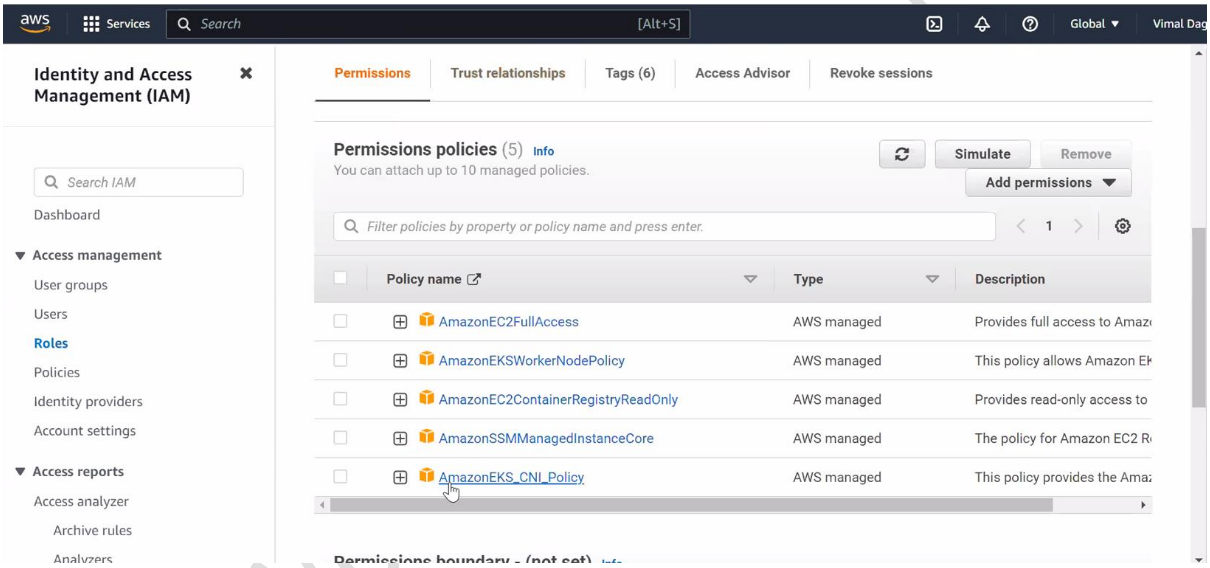
Dashboard
Access management
User groups
Users
Roles
Policies
Identity providers
Account settings
Access reports
Access analyzer
Archive rules
Analyzers
IAM > Roles
Roles (12) Info
An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.
Delete
Create role
eksctl-cluster1-nodegroup-ng-9aa5-NodeInstanceRole-16XLYN14WLI4H
1 match
Role name
Trusted entities
eksctl-cluster1-nodegroup-ng-9aa5-NodeInstanceRole-16XLYN14WLI4H
AWS Service: ec2
Roles Anywhere Info
Authenticate your non AWS workloads and securely provide access to AWS services.
Manage
Access AWS from your non AWS workloads
X.509 Standard
Use your own existing PKI
Temporary credentials
Use temporary credentials with ease



us-east-1.console.aws.amazon.com/iamv2/home?region=ap-south-1#/roles/details/eksctl-cluster1-nodegroup-ng-9aa5-NodeIn...
aws
Services
Search
[Alt+S]
Global
Identity and Access Management (IAM)
Search IAM
Dashboard
Access management
User groups
Users
Roles
Policies
Identity providers
Account settings
Access reports
Access analyzer
Archive rules
Analyzers
Last activity
8 minutes ago
Maximum session duration
1 hour
Permissions
Trust relationships
Tags (6)
Access Advisor
Revoke sessions
Permissions policies (4) Info
You can attach up to 10 managed policies.
Simulate
Remove
Add permissions
Attach policies
Create inline policy
Filter policies by property or policy name and press enter.
Policy name
Type
Description
AmazonEKSWorkerNodePolicy
AWS managed
AmazonEC2ContainerRegistryReadOnly
AWS managed
Provides read-only acc
AmazonSSMManagedInstanceCore
AWS managed
The policy for Amazon
AmazonEKS_CNI_Policy
AWS managed
This policy provides th

[AWS]


us-east-1.console.aws.amazon.com/iamv2/home?region=ap-south-1#/roles/details/eksctl-cluster1-nodegroup-ng-9aa5-NodeIn...
Services
Search
[Alt+S]
Global
AlexaForBusinessGatewayExecution | AWS managed | Provide gateway executi
AmazonElasticTranscoder_ReadOnlyAccess | AWS managed | Grants users read-only a
AmazonRDSFullAccess | AWS managed | Provides full access to A
SupportUser | AWS managed - job function | This policy grants permi
AmazonEC2FullAccess | AWS managed | Provides full access to A
SecretsManagerReadWrite | AWS managed | Provides read/write acce
AWSIoTThingsRegistration | AWS managed | This policy allows users
AmazonDocDBReadOnlyAccess | AWS managed | Provides read-only acce
AmazonMQApiFullAccess | AWS managed | Provides full access to A
AWSElementalMediaStoreReadOnly | AWS managed | Provides read-only perm
Cancel
Add permissions


Services
Search
[Alt+S]
Global
Vimal Dag
Identity and Access Management (IAM)
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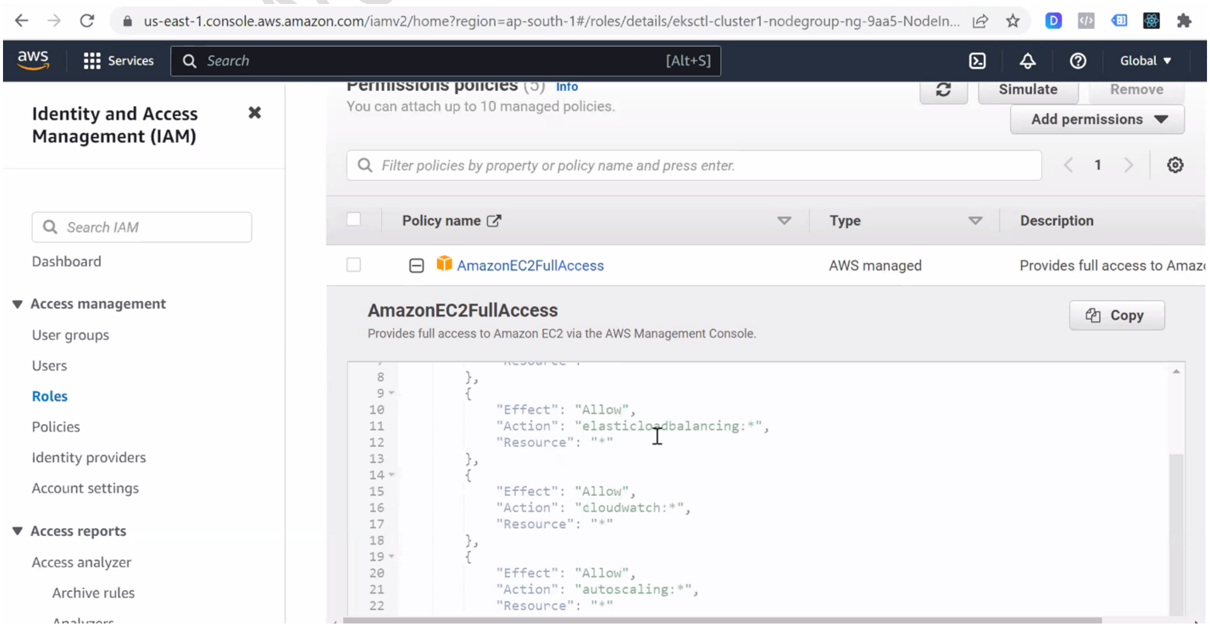
Dashboard
Access management
User groups
Users
Roles
Policies
Identity providers
Account settings
Access reports
Access analyzer
Archive rules
Analyzers
Permissions
Trust relationships
Tags (6)
Access Advisor
Revoke sessions
Permissions policies (5) Info
You can attach up to 10 managed policies.
Simulate
Remove
Add permissions
Filter policies by property or policy name and press enter.
Policy name | Type | Description
AmazonEC2FullAccess | AWS managed | Provides full access to Amaz
AmazonEKSWorkerNodePolicy | AWS managed | This policy allows Amazon EK
AmazonEC2ContainerRegistryReadOnly | AWS managed | Provides read-only access to
AmazonSSMManagedInstanceCore | AWS managed | The policy for Amazon EC2 R
AmazonEKS_CNI_Policy | AWS managed | This policy provides the Amaz


us-east-1.console.aws.amazon.com/iamv2/home?region=ap-south-1#/roles/details/eksctl-cluster1-nodegroup-ng-9aa5-NodeIn...
Services
Search
[Alt+S]
Global
Identity and Access Management (IAM)
Search IAM
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Access management
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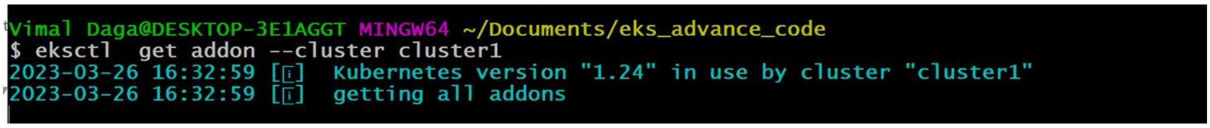
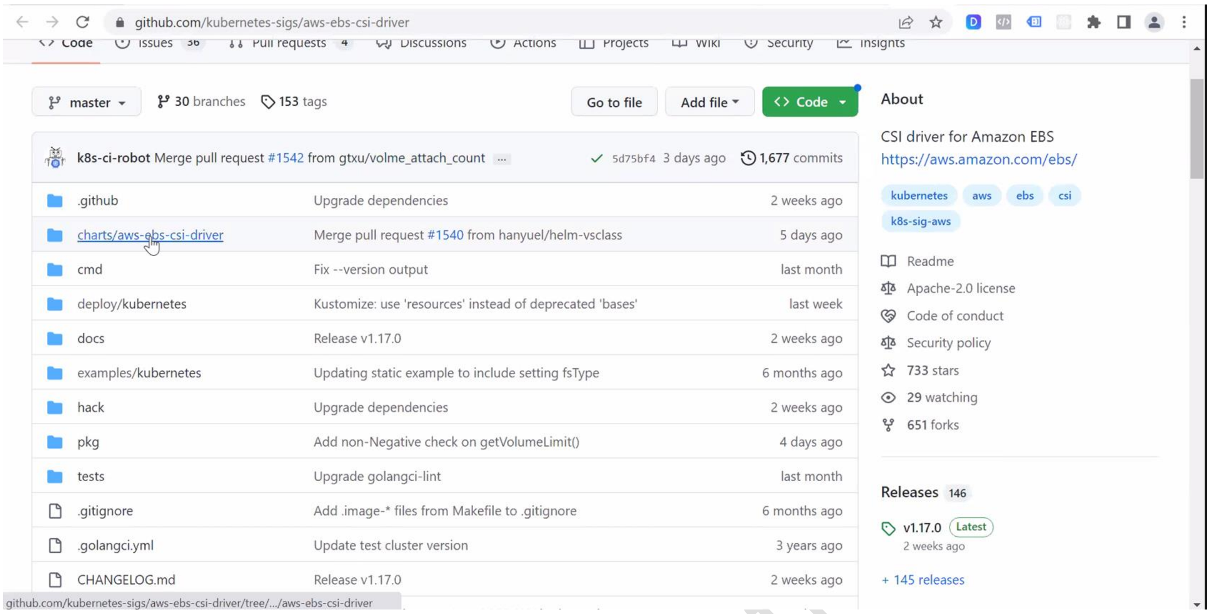
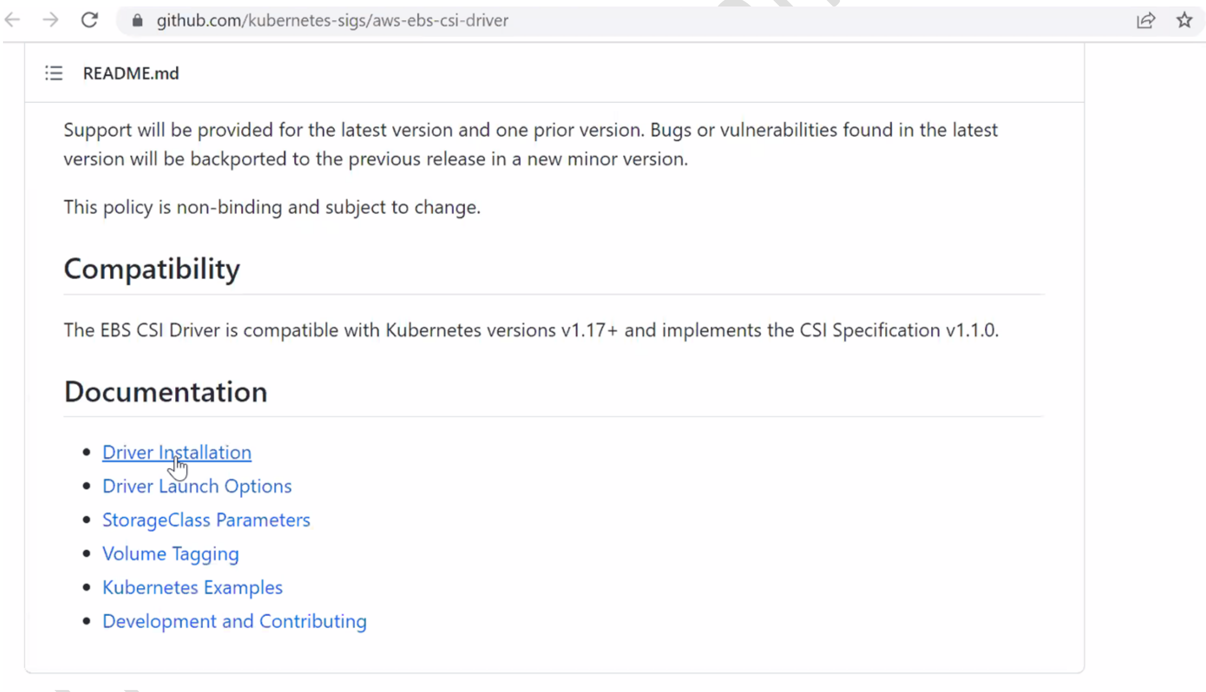
Policies
Identity providers
Account settings
Access reports
Access analyzer
Archive rules
You can attach up to 10 managed policies.
Simulate
Remove
Add permissions
Filter policies by property or policy name and press enter.
Policy name | Type | Description
AmazonEC2FullAccess | AWS managed | Provides full access to Amaz
AmazonEC2FullAccess
Provides full access to Amazon EC2 via the AWS Management Console.
Copy
},
{
"Effect": "Allow",
"Action": "elasticloadbalancing:*",
"Resource": "*"
},
{
"Effect": "Allow",
"Action": "cloudwatch:*",
"Resource": "*"
},
{
"Effect": "Allow",
"Action": "autoscaling:*",
"Resource": "*"

[AWS]


Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ eksctl get addon --cluster cluster1
2023-03-26 16:32:59 [ℹ] Kubernetes version "1.24" in use by cluster "cluster1"
2023-03-26 16:32:59 [ℹ] getting all addons


github.com/kubernetes-sigs/aws-ebs-csi-driver
Code Issues 36 Pull requests 4 Discussions Actions Projects Wiki Security Insights
master 30 branches 153 tags
Go to file Add file Code
About
CSI driver for Amazon EBS
https://aws.amazon.com/ebs/
kubernetes aws ebs csi k8s-sig-aws
k8s-ci-robot Merge pull request #1542 from gtxu/volme_attach_count ... 5d75bf4 3 days ago 1,677 commits
.github Upgrade dependencies 2 weeks ago
charts/aws-ebs-csi-driver Merge pull request #1540 from hanyuel/helm-vsclass 5 days ago
cmd Fix --version output last month
deploy/kubernetes Kustomize: use 'resources' instead of deprecated 'bases' last week
docs Release v1.17.0 2 weeks ago
examples/kubernetes Updating static example to include setting fsType 6 months ago
hack Upgrade dependencies 2 weeks ago
pkg Add non-Negative check on getVolumeLimit() 4 days ago
tests Upgrade golangci-lint last month
.gitignore Add .image-* files from Makefile to .gitignore 6 months ago
.golangci.yml Update test cluster version 3 years ago
CHANGELOG.md Release v1.17.0 2 weeks ago
Readme
Apache-2.0 license
Code of conduct
Security policy
733 stars
29 watching
651 forks
Releases 146
v1.17.0 Latest
2 weeks ago
+ 145 releases
github.com/kubernetes-sigs/aws-ebs-csi-driver/tree/.../aws-ebs-csi-driver


github.com/kubernetes-sigs/aws-ebs-csi-driver
README.md
Support will be provided for the latest version and one prior version. Bugs or vulnerabilities found in the latest version will be backported to the previous release in a new minor version.
This policy is non-binding and subject to change.
Compatibility
The EBS CSI Driver is compatible with Kubernetes versions v1.17+ and implements the CSI Specification v1.1.0.
Documentation
Driver Installation
Driver Launch Options
StorageClass Parameters
Volume Tagging
Kubernetes Examples
Development and Contributing

github.com/kubernetes-sigs/aws-ebs-csi-driver/blob/master/docs/install.md

155 lines (130 sloc) | 6.88 KB

Raw Blame

By default, the driver controller tolerates taint `CriticalAddonsOnly` and has `tolerationSeconds` configured as `300`; and the driver node tolerates all taints. If you don't want to deploy the driver node on all nodes, please set Helm `Value.node.tolerateAllTaints` to false before deployment. Add policies to `Value.node.tolerations` to configure customized toleration for nodes.

Deploy driver

You may deploy the EBS CSI driver via Kustomize, Helm, or as an Amazon EKS managed add-on.

Kustomize

```
kubectl apply -k "github.com/kubernetes-sigs/aws-ebs-csi-driver/deploy/kubernetes/overlays/stable/?ref=release-1.17"
```

Note: Using the master branch to deploy the driver is not supported as the master branch may contain upcoming features incompatible with the currently released stable version of the driver.

Helm

- Add the `aws-ebs-csi-driver` Helm repository.

```
helm repo add aws-ebs-csi-driver https://kubernetes-sigs.github.io/aws-ebs-csi-driver
helm repo update
```

```
$

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl apply -k "github.com/kubernetes-sigs/aws-ebs-csi-driver/deploy/kubernetes/overlays/stable
/?ref=release-1.17"   --kubeconfig  kube-cluster1.config
serviceaccount/ebs-csi-controller-sa created
serviceaccount/ebs-csi-node-sa created
clusterrole.rbac.authorization.k8s.io/ebs-csi-node-role created
clusterrole.rbac.authorization.k8s.io/ebs-external-attacher-role created
clusterrole.rbac.authorization.k8s.io/ebs-external-provisioner-role created
clusterrole.rbac.authorization.k8s.io/ebs-external-resizer-role created
clusterrole.rbac.authorization.k8s.io/ebs-external-snapshotter-role created
clusterrolebinding.rbac.authorization.k8s.io/ebs-csi-attacher-binding created
clusterrolebinding.rbac.authorization.k8s.io/ebs-csi-node-getter-binding created
clusterrolebinding.rbac.authorization.k8s.io/ebs-csi-provisioner-binding created
clusterrolebinding.rbac.authorization.k8s.io/ebs-csi-resizer-binding created
clusterrolebinding.rbac.authorization.k8s.io/ebs-csi-snapshotter-binding created
deployment.apps/ebs-csi-controller created
poddisruptionbudget.policy/ebs-csi-controller created
daemonset.apps/ebs-csi-node created
csidriver.storage.k8s.io/ebs.csi.aws.com created

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl  get pods  -n kube-system    --kubeconfig  kube-cluster1.config
NAME                                  READY   STATUS    RESTARTS   AGE
aws-node-418xs                        1/1     Running   0          49m
aws-node-qqg7w                        1/1     Running   0          49m
coredns-74ccfbd98b-45w5p              1/1     Running   0          58m
coredns-74ccfbd98b-5826h              1/1     Running   0          58m
ebs-csi-controller-5b769c467d-h8tmx   6/6     Running   0          34s
ebs-csi-controller-5b769c467d-pbbsf   6/6     Running   0          34s
ebs-csi-node-9fm47                    3/3     Running   0          34s
ebs-csi-node-frv2v                    3/3     Running   0          34s
kube-proxy-6jcll                      1/1     Running   0          49m
kube-proxy-tkbfd                      1/1     Running   0          49m

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$
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl  get  sc  --kubeconfig  kube-cluster1.config
NAME            PROVISIONER             RECLAIMPOLICY   VOLUMEBINDINGMODE      ALLOWVOLUMEEXPANSION
   AGE
gp2 (default)   kubernetes.io/aws-ebs   Delete          WaitForFirstConsumer   false
   60m
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl  describe   sc  gp2  --kubeconfig  kube-cluster1.config
Name:            gp2
IsDefaultClass:  Yes
Annotations:     kubectl.kubernetes.io/last-applied-configuration={"apiVersion":"storage.k8s.io/v1"
,"kind":"StorageClass","metadata":{"annotations":{"storageclass.kubernetes.io/is-default-class":"tr
ue"},"name":"gp2"},"parameters":{"fsType":"ext4","type":"gp2"},"provisioner":"kubernetes.io/aws-ebs
","volumeBindingMode":"WaitForFirstConsumer"}
,storageclass.kubernetes.io/is-default-class=true
Provisioner:           kubernetes.io/aws-ebs
Parameters:            fsType=ext4,type=gp2
AllowVolumeExpansion:  <unset>
MountOptions:          <none>
ReclaimPolicy:         Delete
VolumeBindingMode:     WaitForFirstConsumer
Events:                <none>
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl get pods -n kube-system -l app.kubernetes.io/name=aws-ebs-csi-driver  --kubeconfig  kube-
cluster1.config
NAME                                  READY   STATUS    RESTARTS   AGE
ebs-csi-controller-5b769c467d-h8tmx   6/6     Running   0          4m20s
ebs-csi-controller-5b769c467d-pbbsf   6/6     Running   0          4m20s
ebs-csi-node-9fm47                    3/3     Running   0          4m20s
ebs-csi-node-frv2v                    3/3     Running   0          4m20s
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl apply  -f https://raw.githubusercontent.com/vimallinuxworld13/aws_eks_code/master/storage
_class_add_for_ebs_csi  --kubeconfig  kube-cluster1.config
storageclass.storage.k8s.io/ebs-sc created

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl  get  sc  --kubeconfig  kube-cluster1.config
NAME            PROVISIONER             RECLAIMPOLICY   VOLUMEBINDINGMODE      ALLOWVOLUMEEXPANSION
   AGE
ebs-sc          ebs.csi.aws.com         Delete          WaitForFirstConsumer   false
   7s
gp2 (default)   kubernetes.io/aws-ebs   Delete          WaitForFirstConsumer   false
   64m

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ |
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl  describe   sc  ebs-sc  --kubeconfig  kube-cluster1.config
Name:            ebs-sc
IsDefaultClass:  No
Annotations:     kubectl.kubernetes.io/last-applied-configuration={"apiVersion":"storage.k8s.io/v1"
,"kind":"StorageClass","metadata":{"annotations":{},"name":"ebs-sc"},"provisioner":"ebs.csi.aws.com
","volumeBindingMode":"WaitForFirstConsumer"}

Provisioner:           ebs.csi.aws.com
Parameters:            <none>
AllowVolumeExpansion:  <unset>
MountOptions:          <none>
ReclaimPolicy:         Delete
VolumeBindingMode:     WaitForFirstConsumer
Events:                <none>
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl  describe   sc  gp2  --kubeconfig  kube-cluster1.config
Name:            gp2
IsDefaultClass:  Yes
Annotations:     kubectl.kubernetes.io/last-applied-configuration={"apiVersion":"storage.k8s.io/v1"
,"kind":"StorageClass","metadata":{"annotations":{"storageclass.kubernetes.io/is-default-class":"tr
ue"},"name":"gp2"},"parameters":{"fsType":"ext4","type":"gp2"},"provisioner":"kubernetes.io/aws-ebs
","volumeBindingMode":"WaitForFirstConsumer"}
,storageclass.kubernetes.io/is-default-class=true
Provisioner:           kubernetes.io/aws-ebs
Parameters:            fsType=ext4,type=gp2
AllowVolumeExpansion:  <unset>
MountOptions:          <none>
ReclaimPolicy:         Delete
VolumeBindingMode:     WaitForFirstConsumer
Events:                <none>
```

```
Vimal Daga@DESKTOP-
$ kubectl  describe
Name:           gp
IsDefaultClass:  Ye
Annotations:     ku
,"kind":"StorageCla
ue"},"name":"gp2"},
","volumeBindingMod
,storageclass.kuber
Provisioner:
Parameters:
AllowVolumeExpansio
MountOptions:
ReclaimPolicy:
VolumeBindingMode:
Events:

Vimal Daga@DESKTOP-
$ kubectl  edit
Edit cancelled, no

Vimal Daga@DESKTOP-
$ kubectl  edit
```

File Edit Format View Help

```
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
  annotations:
    kubectl.kubernetes.io/last-applied-configuration: |
      {"apiVersion":"storage.k8s.io/v1","kind":"StorageClas
  creationTimestamp: "2023-03-26T11:13:17Z"
  name: ebs-sc
  resourceVersion: "11157"
  uid: 2f7bfe25-6944-4035-90b9-17991496a1fc
parameters:
  fsType: ext4
  type: gp2
provisioner: ebs.csi.aws.com
reclaimPolicy: Delete
volumeBindingMode: WaitForFirstConsumer
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl  edit    sc  gp2  --kubeconfig  kube-cluster1.config
Edit cancelled, no changes made.

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl  edit    sc  ebs-sc  --kubeconfig  kube-cluster1.config
error: storageclasses.storage.k8s.io "ebs-sc" is invalid
A copy of your changes has been stored to "C:\\Users\\VIMALD~1\\AppData\\Local\\Temp\\kub
dit-2629302386.yaml"
error: Edit cancelled, no valid changes were saved.

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl  delete    sc  ebs-sc  --kubeconfig  kube-cluster1.config
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ vim mysc.yml
```

```
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
  name: ebs-sc
provisioner: ebs.csi.aws.com
volumeBindingMode: WaitForFirstConsumer
parameters:
  type: gp3
  fsType: ext4
~
mysc.yml[+] [unix] (05:29 01/01/1970)                                9,15 All
-- INSERT --
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl apply  -f mysc.yml  --kubeconfig  kube-cluster1.config
storageclass.storage.k8s.io/ebs-sc created

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$
```

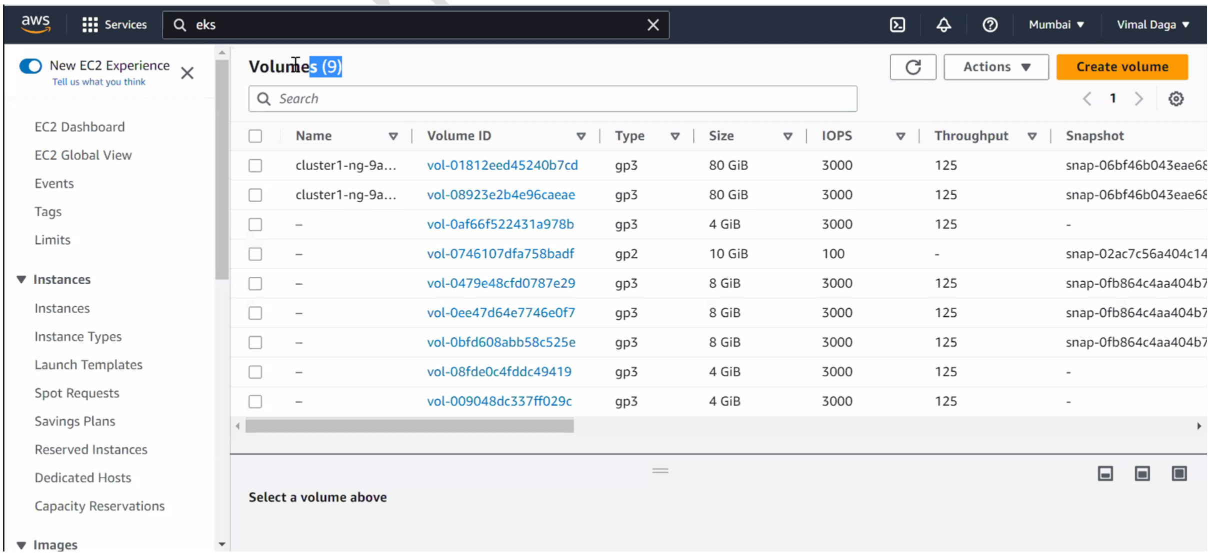
[AWS]

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl  describe   sc  ebs-sc  --kubeconfig  kube-cluster1.config
Name:            ebs-sc
IsDefaultClass:  No
Annotations:     kubectl.kubernetes.io/last-applied-configuration={"apiVersion":"storage.k8s.io/v1"
,"kind":"StorageClass","metadata":{"annotations":{},"name":"ebs-sc"},"parameters":{"fsType":"ext4",
"type":"gp3"},"provisioner":"ebs.csi.aws.com","volumeBindingMode":"WaitForFirstConsumer"}

Provisioner:           ebs.csi.aws.com
Parameters:            fsType=ext4,type=gp3
AllowVolumeExpansion:  <unset>
MountOptions:          <none>
ReclaimPolicy:         Delete
VolumeBindingMode:     WaitForFirstConsumer
Events:                <none>
```

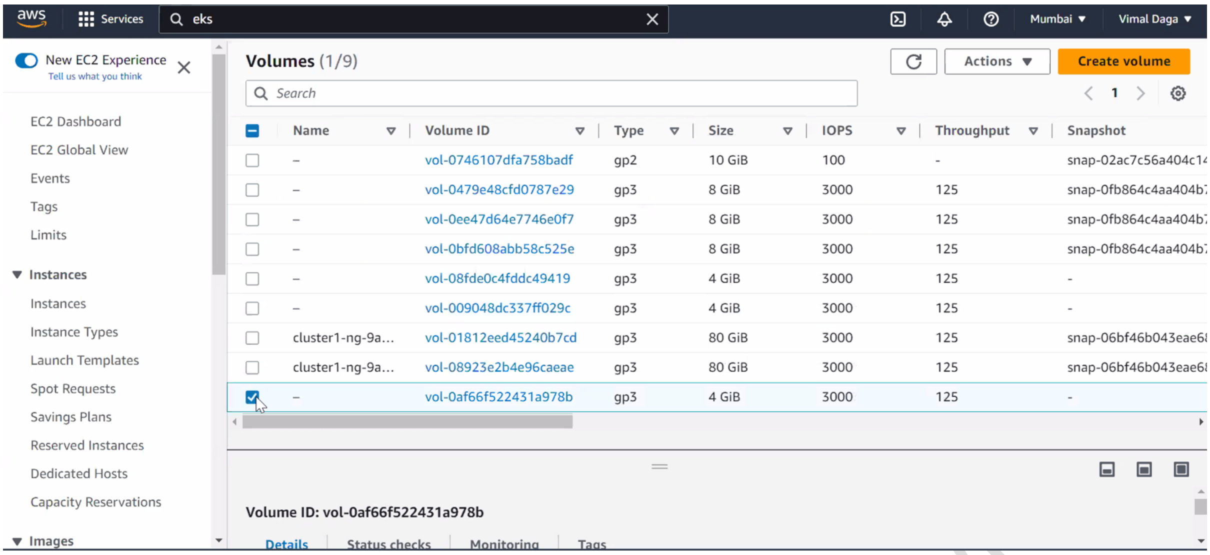
```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl apply -f https://raw.githubusercontent.com/vimallinuxworld13/aws_eks_code/master/PVC_ebs_
sc  --kubeconfig  kube-cluster1.config
persistentvolumeclaim/ebs-mysql-pv-claim created
```

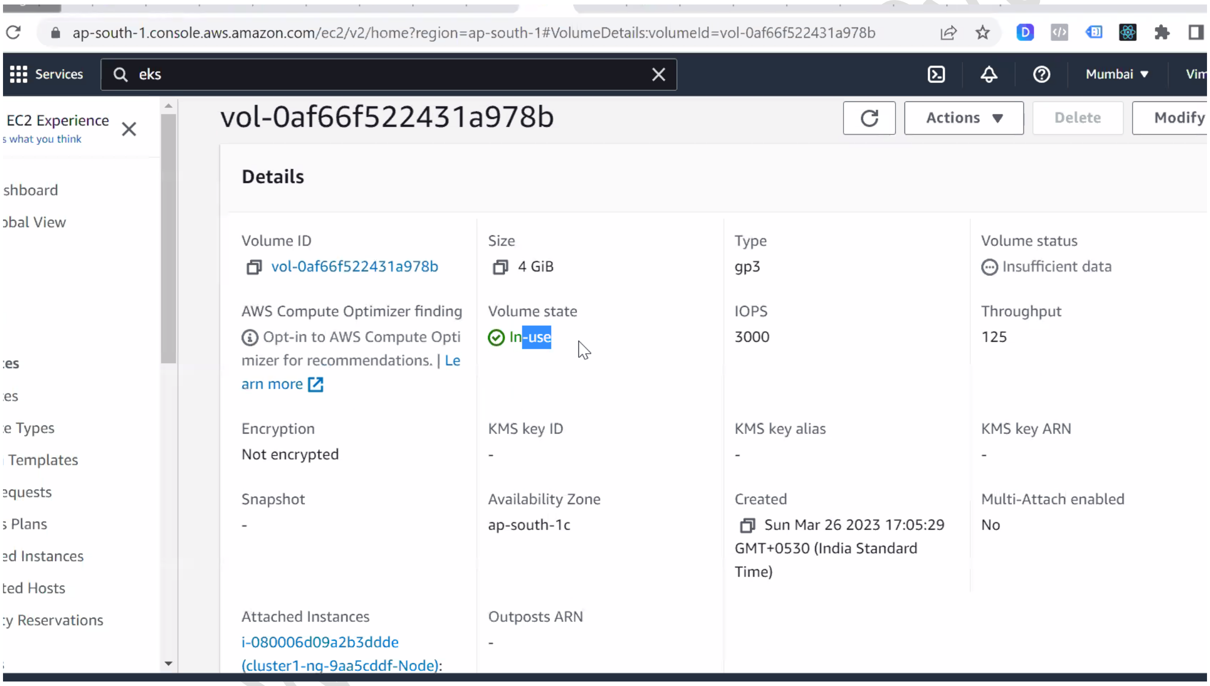
```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl   get pvc  --kubeconfig  kube-cluster1.config
NAME                 STATUS    VOLUME    CAPACITY    ACCESS MODES    STORAGECLASS    AGE
ebs-mysql-pv-claim   Pending                                         ebs-sc          3m29s

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl   apply -f  https://raw.githubusercontent.com/vimallinuxworld13/aws_eks_code/master/mysql
_db_secret.yml    --kubeconfig  kube-cluster1.config
secret/mysecure created

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl   get secrets   --kubeconfig  kube-cluster1.config
NAME       TYPE     DATA   AGE
mysecure   Opaque   2      8s

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl   apply -f https://raw.githubusercontent.com/vimallinuxworld13/aws_eks_code/master/mysql_
deploy_pvc_ebs.yml     --kubeconfig  kube-cluster1.config
pod/dbpod1 created

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl   get pvc  --kubeconfig  kube-cluster1.config
NAME                 STATUS   VOLUME                                     CAPACITY   ACCESS MODES
STORAGECLASS   AGE
ebs-mysql-pv-claim   Bound    pvc-8b96cd8e-06ed-4108-a658-6661f16a6b72   4Gi        RWO
ebs-sc         10m
```

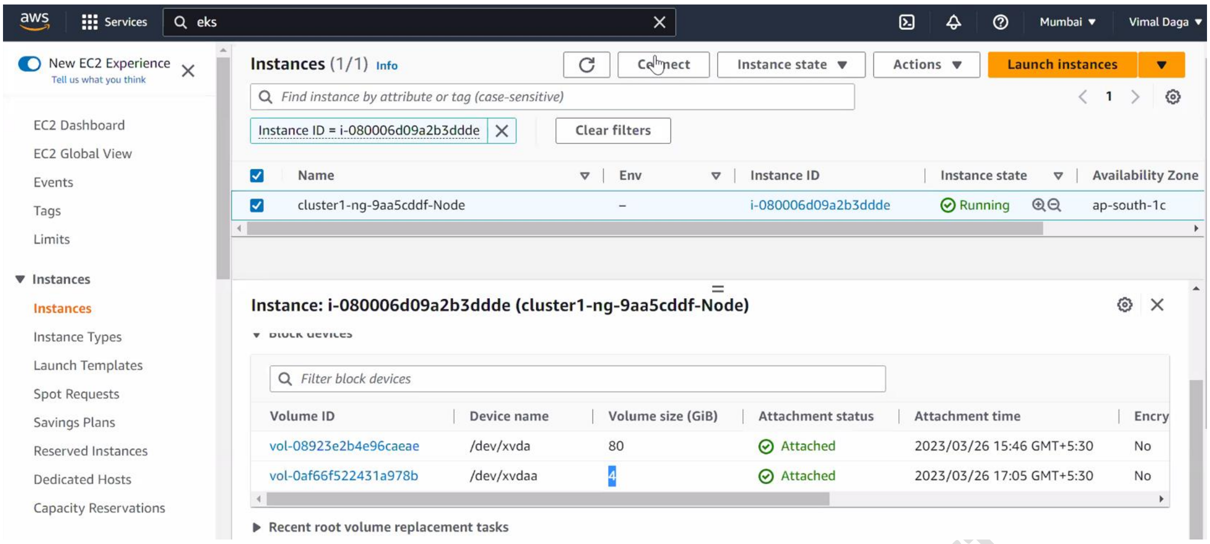


[AWS]






```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl   get pvc  --kubeconfig  kube-cluster1.config
NAME                STATUS    VOLUME                                     CAPACITY   ACCESS MODES
STORAGECLASS   AGE
ebs-mysql-pv-claim   Bound    pvc-8b96cd8e-06ed-4108-a658-6661f16a6b72   4Gi        RWO
ebs-sc         10m

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl   get pv  --kubeconfig  kube-cluster1.config
NAME                                       CAPACITY   ACCESS MODES   RECLAIM POLICY   STATUS   CLAI
M                         STORAGECLASS   REASON   AGE
pvc-8b96cd8e-06ed-4108-a658-6661f16a6b72   4Gi        RWO            Delete           Bound    defa
ult/ebs-mysql-pv-claim   ebs-sc                   9s

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl   get pods  --kubeconfig  kube-cluster1.config
NAME      READY   STATUS    RESTARTS   AGE
dbpod1    1/1     Running   0          2m39s

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ 
```




```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl   get pods  --kubeconfig  kube-cluster1.config
NAME     READY   STATUS    RESTARTS   AGE
dbpod1   1/1     Running   0          5m54s
```

```
Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl  --kubeconfig  kube-cluster1.config   exec  dbpod1  date
kubectl exec [POD] [COMMAND] is DEPRECATED and will be removed in a future version. Use kubectl exe
c [POD] -- [COMMAND] instead.
Sun Mar 26 11:43:10 UTC 2023

Vimal Daga@DESKTOP-3E1AGGT MINGW64 ~/Documents/eks_advance_code
$ kubectl  --kubeconfig  kube-cluster1.config   exec  dbpod1  --  date
Sun Mar 26 11:43:15 UTC 2023
```

- Pod becomes consumer of PVC.
- 1 EBS can connect to only a single pod.
- Multi AZ is not supported by EBS so we can use EFS for this kind of usecase.
- Annotations is for machine It is used as a kind of metadata, k8s gets lots of information from annotation.

Github Link-

https://github.com/vimallinuxworld13/aws_eks_code